Exp. No: 12

Verification of Poynting Theorem and Determination of Energy Flow Using the Poynting Vector

Aim

To verify the Poynting theorem and to determine the instantaneous and time-average power density (Poynting vector) of a uniform plane wave in free space and in a lossy medium, of a current-carrying conductor, and of solar radiation incident on a panel.

Objectives

1. To determine the instantaneous and average Poynting vector of a uniform plane wave and verify the point-form Poynting theorem numerically.

2. To calculate the attenuation, intrinsic impedance and power density in a lossy medium and show that the power entering equals the power dissipated.

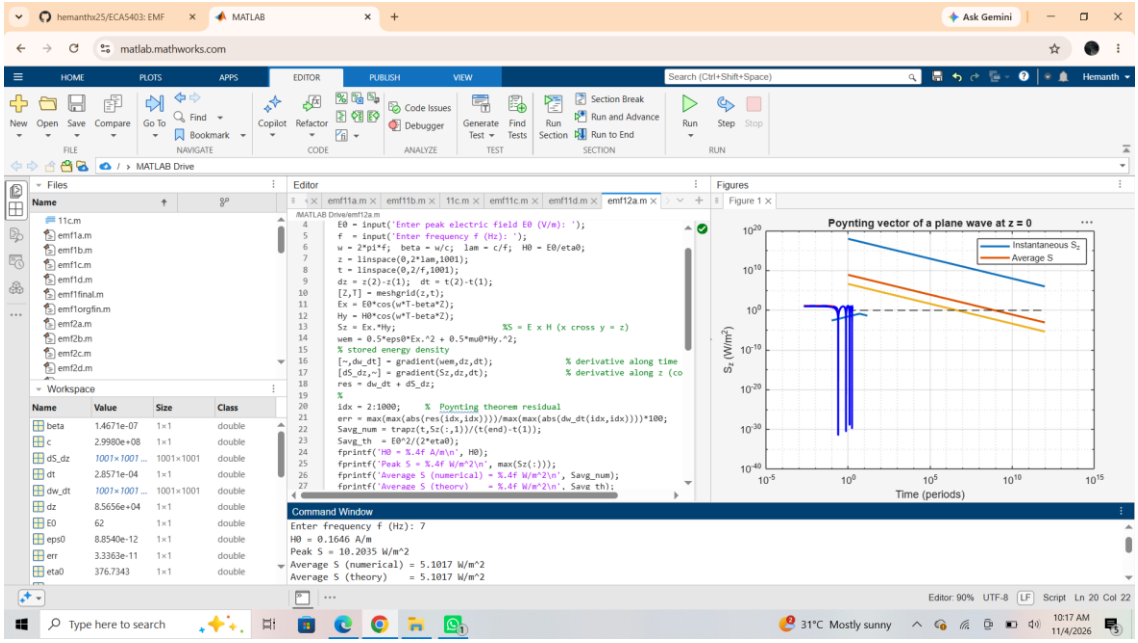
3. To show that the Poynting vector flux into a current-carrying conductor equals the Joule loss I^2R.

4. To relate solar irradiance to the field amplitudes and evaluate the power received by a panel at different angles of incidence.

Test Case (i) – Poynting Vector and Poynting Theorem for a Plane Wave in Free Space

```
mu0 = 4*pi*1e-7;  eps0 = 8.854e-12;
c = 1/sqrt(mu0*eps0);  eta0 = sqrt(mu0/eps0);
E0 = input('Enter peak electric field E0 (V/m): ');
f  = input('Enter frequency f (Hz): ');
w = 2*pi*f;  beta = w/c;  lam = c/f;  H0 = E0/eta0;
z = linspace(0,2*lam,1001);
t = linspace(0,2/f,1001);
dz = z(2)-z(1);  dt = t(2)-t(1);
[Z,T] = meshgrid(z,t);
Ex = E0*cos(w*T-beta*Z);
Hy = H0*cos(w*T-beta*Z);
Sz = Ex.*Hy;                          % S = E x H (x cross y = z)
wem = 0.5*eps0*Ex.^2 + 0.5*mu0*Hy.^2;         % stored energy density
[~,dw_dt] = gradient(wem,dz,dt);              % derivative along time (rows)
[dS_dz,~] = gradient(Sz,dz,dt);               % derivative along z (columns)
res = dw_dt + dS_dz;                          % Poynting theorem residual
idx = 2:1000;
err = max(max(abs(res(idx,idx))))/max(max(abs(dw_dt(idx,idx))))*100;
Savg_num = trapz(t,Sz(:,1))/(t(end)-t(1));
Savg_th  = E0^2/(2*eta0);
fprintf('H0 = %.4f A/m\n', H0);
fprintf('Peak S = %.4f W/m^2\n', max(Sz(:)));
fprintf('Average S (numerical) = %.4f W/m^2\n', Savg_num);
fprintf('Average S (theory)    = %.4f W/m^2\n', Savg_th);
fprintf('Poynting theorem residual = %.3e %%\n', err);
plot(t*f,Sz(:,1),'b','LineWidth',2); hold on;
plot([0 2],[Savg_th Savg_th],'r--','LineWidth',1.5); grid on;
xlabel('Time (periods)'); ylabel('S_z (W/m^2)');
legend('Instantaneous S_z','Average S');
title('Poynting vector of a plane wave at z = 0');
```

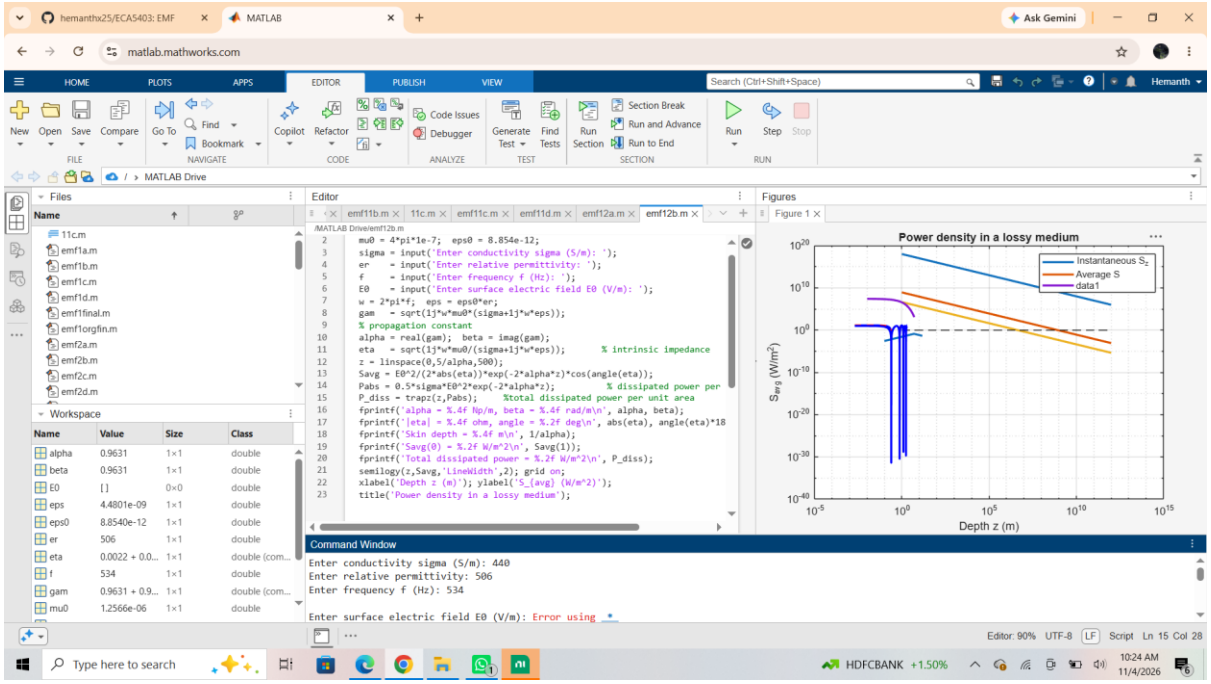
Output:



Test Case (ii) – Energy Flow and Power Absorption in a Lossy Medium (Sea Water)

```
clc; clear;
mu0 = 4*pi*1e-7;  eps0 = 8.854e-12;
sigma = input('Enter conductivity sigma (S/m): ');
er    = input('Enter relative permittivity: ');
f     = input('Enter frequency f (Hz): ');
E0    = input('Enter surface electric field E0 (V/m): ');
w = 2*pi*f;  eps = eps0*er;
gam   = sqrt(1j*w*mu0*(sigma+1j*w*eps));      % propagation constant
alpha = real(gam);  beta = imag(gam);
eta   = sqrt(1j*w*mu0/(sigma+1j*w*eps));      % intrinsic impedance
z = linspace(0,5/alpha,500);
Savg = E0^2/(2*abs(eta))*exp(-2*alpha*z)*cos(angle(eta));
Pabs = 0.5*sigma*E0^2*exp(-2*alpha*z);        % dissipated power per unit volume
P_diss = trapz(z,Pabs);                       % total dissipated power per unit area
fprintf('alpha = %.4f Np/m, beta = %.4f rad/m\n', alpha, beta);
fprintf('|eta| = %.4f ohm, angle = %.2f deg\n', abs(eta), angle(eta)*180/pi);
fprintf('Skin depth = %.4f m\n', 1/alpha);
fprintf('Savg(0) = %.2f W/m^2\n', Savg(1));
fprintf('Total dissipated power = %.2f W/m^2\n', P_diss);
semilogy(z,Savg,'LineWidth',2); grid on;
xlabel('Depth z (m)'); ylabel('S_{avg} (W/m^2)');
title('Power density in a lossy medium');
```

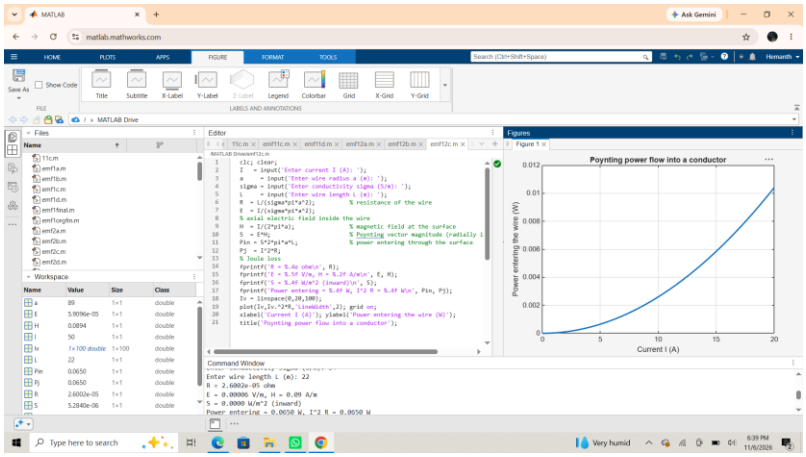
Output:



Test Case (iii) – Poynting Theorem for a Current-Carrying Conductor

```
clc; clear;
I     = input('Enter current I (A): ');
a     = input('Enter wire radius a (m): ');
sigma = input('Enter conductivity sigma (S/m): ');
L     = input('Enter wire length L (m): ');
R  = L/(sigma*pi*a^2);          % resistance of the wire
E  = I/(sigma*pi*a^2);          % axial electric field inside the wire
H  = I/(2*pi*a);                % magnetic field at the surface
S  = E*H;                       % Poynting vector magnitude (radially inward)
Pin = S*2*pi*a*L;               % power entering through the surface
Pj  = I^2*R;                    % Joule loss
fprintf('R = %.4e ohm\n', R);
fprintf('E = %.5f V/m, H = %.2f A/m\n', E, H);
fprintf('S = %.4f W/m^2 (inward)\n', S);
fprintf('Power entering = %.4f W, I^2 R = %.4f W\n', Pin, Pj);
Iv = linspace(0,20,100);
plot(Iv,Iv.^2*R,'LineWidth',2); grid on;
xlabel('Current I (A)'); ylabel('Power entering the wire (W)');
title('Poynting power flow into a conductor');
```

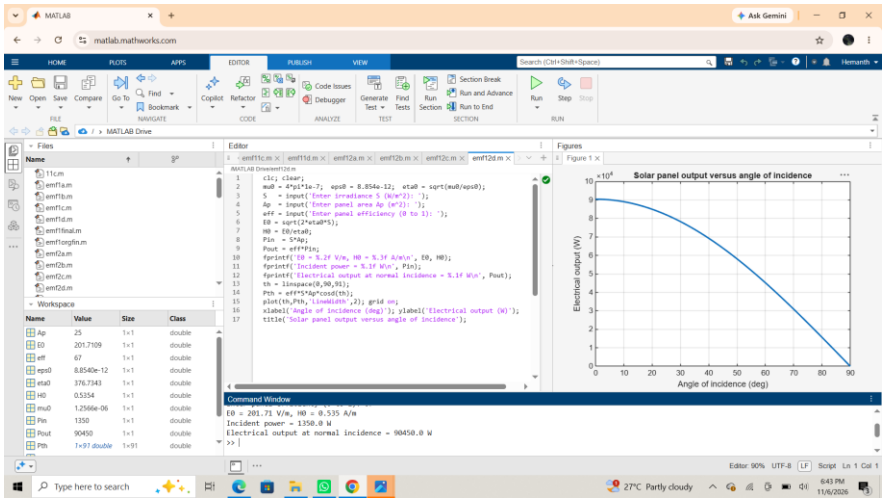
Output:



Test Case (iv) – Poynting Vector Analysis for a Solar Panel (IEC 60904 Irradiance)

```
clc; clear;
mu0 = 4*pi*1e-7;  eps0 = 8.854e-12;  eta0 = sqrt(mu0/eps0);
S   = input('Enter irradiance S (W/m^2): ');
Ap  = input('Enter panel area Ap (m^2): ');
eff = input('Enter panel efficiency (0 to 1): ');
E0 = sqrt(2*eta0*S);
H0 = E0/eta0;
Pin  = S*Ap;
Pout = eff*Pin;
fprintf('E0 = %.2f V/m, H0 = %.3f A/m\n', E0, H0);
fprintf('Incident power = %.1f W\n', Pin);
fprintf('Electrical output at normal incidence = %.1f W\n', Pout);
th = linspace(0,90,91);
Pth = eff*S*Ap*cosd(th);
plot(th,Pth,'LineWidth',2); grid on;
xlabel('Angle of incidence (deg)'); ylabel('Electrical output (W)');
title('Solar panel output versus angle of incidence');
```

Output:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify E and H vectors and coordinate orientation	Identify E and H vectors and coordinate orientation	Correctly specifies field vectors, directions, units and applicable assumptions	Most vectors and units are correct.	Some directions or units need correction.	Vector setup is incorrect.	25	2.5	
2	Calculate Poynting vector and power	Calculate Poynting vector and power	Computes S = E × H with correct direction and units; integrates over a surface when required.	Calculations are mostly correct with minor errors.	Partial calculations need assistance.	Power-flow calculation is incorrect.	25	2.5	
3	Verify energy flow result	Verify energy flow result	Checks vector direction, magnitude and energy/power relation against theory or plotted fields	Verification is mostly correct.	Verification is incomplete.	No valid check is provided.	30	3	
4	Interpret and report findings	Interpret and report findings	Explains direction and meaning of electromagnetic energy flow and documents results clearly.	Explanations is sound with minor omissions.	Interpretation is limited.	Results are not interpreted or documented.	20	2	
Total							100	10	